2020春高数下试题2 (回忆版)

1. Determine whether the following statements are true or false? No justification is necessary.

(1) Parametric curves $x(t)=\cos t$, $y(t)=\sin t$ and $x(t)=\sin t$, $y(t)=\cos t$ have the same graph.

(2) If $x(t)=f(t)$ and $y(t)=g(t)$ are twice differentiable, then

$$\frac{d^2y}{dx^2}=\frac{d^2f(t)/dt^2}{d^2g(t)/dt^2}$$

2. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) If $|u|=2$, $|v|=\sqrt{2}$, and $u\cdot v=2$, then $|u\times v|$ is

(A) 2 (B) $2\sqrt{2}$ (C) $\frac{\sqrt{2}}{2}$ (D) 1

(2) How many points of intersection do the curves $r=\frac{1}{2}$ and $r=\cos 2\theta$ have?

(A) 2 (B) 4 (C) 6 (D) 8

(3) If $f(x+y,x-y)=x^2-y^2$, then $\frac{\partial f(x,y)}{\partial x}+\frac{\partial f(x,y)}{\partial y}=$

(A) $2x-2y$ (B) $2x+2y$ (C) $x-y$ (D) $x+y$.

3. Please fill in the blank for the questions below.

(1) If the plane $3x+\lambda y-3z+16=0$ is tangent to the surface $3x^2+y^2+z^2=16$, then $\lambda=$ _____.

(2) Let $z=\ln\sqrt{x^2+y^2}+\tan^{-1}\frac{x+y}{x-y}$, then $dz=$ _____.

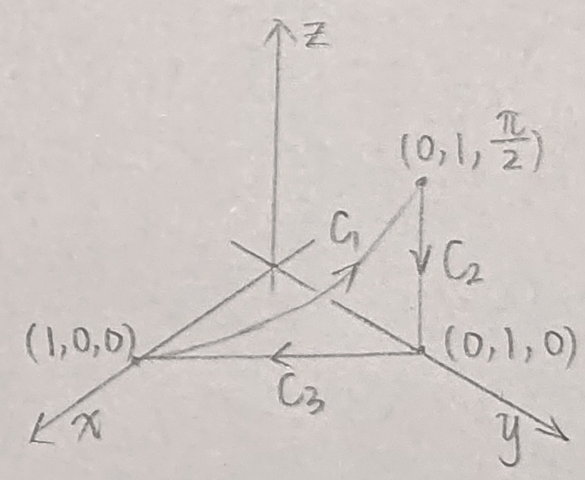
(3) The distance from the point $P(1,4,0)$ to the plane through $A(0,0,0)$, $B(2,0,1)$ and $C(2,-1,0)$ is _____.

(4) A closed path C consists of three curves:

C_1: $r(t)=(\cos t)\,i+(\sin t)\,j+tk$, $0\le t\le\frac{\pi}{2}$

C_2: $r(t)=i+(\frac{\pi}{2})(1-t)\,k$, $\le t\le 1$

C_3: $r(t)=t\,i+(1-t)\,j$, $0\le t\le 1$.



Then the circulation of $F=2x\,i+2z\,j+2y\,k$ around path C traversed in the direction of increasing is _____.

4. Determine the length of polar curve $r=\sin^3(\frac{\theta}{3})$, $0\le\theta\le\frac{\pi}{4}$.

5. Given a curve $r(t)=(\cos^3 t,\sin^3 t,0)$, $0<t<\frac{\pi}{2}$ in R^3, find its curvature and principal unit normal.

6. The sequence $\{a_n\}$ is defined by $a_{2k-1}=\frac{1}{k}$, $a_{2k}=-\frac{1}{k+2}$ (k can be any positive integer). Is the series $\sum_{n=1}^{\infty}a_n$ convergent or divergent? Prove your conclusion.

7. Find the Maclaurin series for $f(x)=\frac{1}{(1+x)^3}$.

8. Find the absolute maximum and minimum values of $f(x,y)=4xy^2-x^2y^2-xy^3$ on the close triangular region in the xy-plane with verties $(0,0)$, $(0,6)$ and $(6,0)$.

9. Find the centroid of the solid bounded above by the surface $z=\sqrt{r}$, on the sides by the cylinder $r=4$, and below by the xy-plane.

10. Use the Stokes' Theorem to compute the surface integral $\iint_S \nabla \times F \cdot n \, d\sigma$, here $F = xz\,i + yz\,j + xy\,k$, and S is the part of the ~~surface~~ sphere $x^2+y^2+z^2=4$ that lies inside the cylinder $x^2+y^2=1$ and above the xy-plane (the boundary is ~~con~~ counterclockwise when viewed from above).

11. Use the Divergence Theorem to find the outward flux of F across the boundary of the region D, here $F = xy\,i + (y^2+e^{xz^2})\,j + \sin(xy)\,k$; and D is the region bounded by the parabolic cylinder $z=1-x^2$, and the planes $z=0$, $y=0$, and $y+z=2$.

